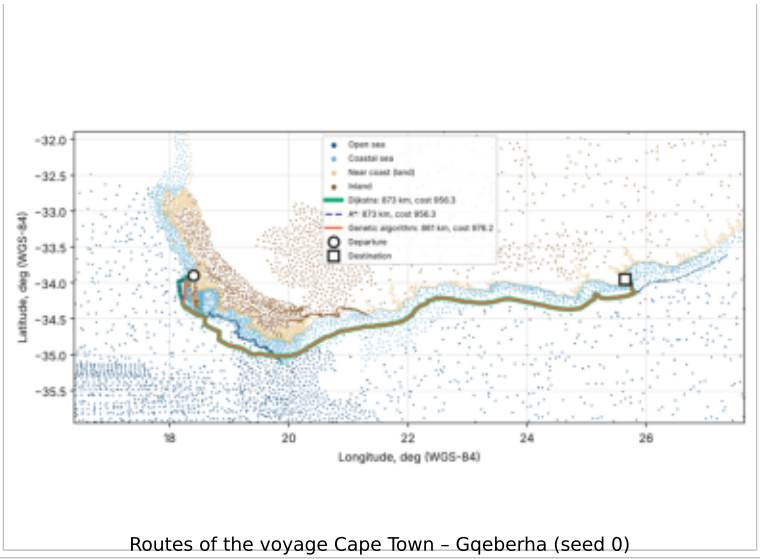
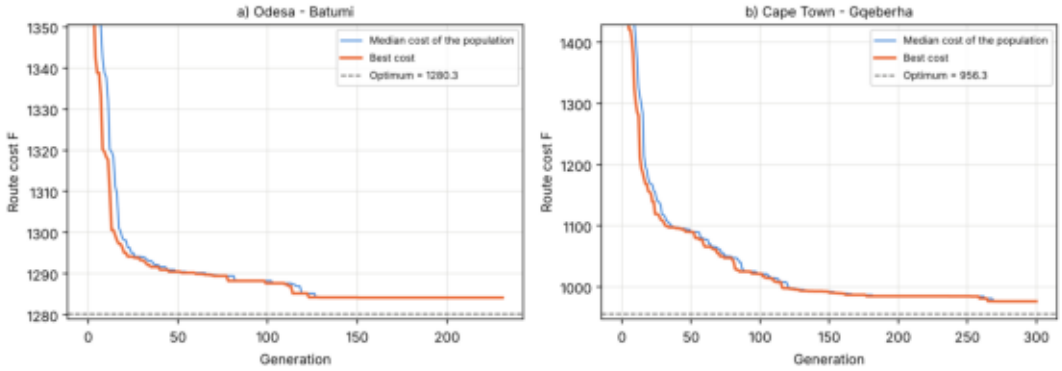
Test results of the application



Routes of the voyage Cape Town – Gqeberha (seed 0)



Convergence of the GA: Odesa – Batumi, Cape Town – Gqeberha

Voyage	F*	Exp. Dijkstra / A*	Time D / A*, ms	GA mean gap, %	GA ≤ 1 %	GA time, s
Odesa – Batumi	1280.3	5940 / 3465	31 / 27	0.28	10 of 10	4.2
Varna – Sinop	836.4	4478 / 2483	18 / 15	2.07	7 of 10	2.2
Burgas – Trabzon	1220.2	5155 / 4113	24 / 31	0.16	10 of 10	3.1
Sanya – Muara	1636.0	8688 / 1926	48 / 13	0.28	10 of 10	3.1
Cape Town – Gqeberha	956.3	8007 / 7279	38 / 49	2.85	0 of 10	5.7
Dubai – Muscat	694.6	2804 / 1289	12 / 8	0.28	10 of 10	3.3
Colon – Limon	626.0	2319 / 1885	10 / 11	0.37	10 of 10	2.5

– Dijkstra and A* give the same optimum; A* expands on average 1.97 times fewer nodes;

– the GA found valid routes in all 70 runs; on 5 of 7 voyages every run was within 1 % of the optimum;

– on the graph built by distance the optimal route crossed land on 2 of 7 voyages;

– without intermediate nodes in the random walks the mean gap on Cape Town – Gqeberha grows from 2.63 % to 12.50 %.

Rev.	Sheet	Doc. No.	Sign.	Date
Developed		Li H.		
Checked		Kozinsky A.A.		
T. contr.				
N. contr.				
Approved		Golovko V.A.		

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Development of an application for finding the optimal course of a sea vessel over open-sea points based on a genetic algorithm

Test results (drawing)

Lit.	Mass	Scale
	–	–

Sheet 5	Sheets 5

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group ИИ-2025